

SHREYAS KUMAR

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PROFESSIONAL SUMMARY

Full-Stack Developer & AI/ML Researcher with expertise in React Native, Next.js, and Python. Published researcher with 92.73% accuracy in medical AI (Nature Portfolio). 3 Published Patents. 2nd Place - Tata Technologies InnoVent 2026 (2,822 teams, 10,247 innovators). IEI Student Chapter Member. Proven track record in building scalable applications and leading technical teams.

EDUCATION

Vellore Institute of Technology

Bachelor of Engineering in Artificial Intelligence | GPA: 9.38/10.0

Chennai, India

Aug 2023 – May 2027

- **IEI Student Chapter Member** - The Institution of Engineers (India) | Membership ID: IEIESC/600127/VIT/CS/04874

TECHNICAL SKILLS

Languages: Python • JavaScript • TypeScript • Java • C++ • Kotlin • SQL • C#

Frontend: React.js • Next.js • React Native • TypeScript • Tailwind CSS • Redux • React Query

Backend & Databases: Node.js • Express.js • Firebase • MongoDB • PostgreSQL • Appwrite

AI/ML & Data Science: TensorFlow • PyTorch • CNN • BiLSTM • Scikit-learn • Pandas • NumPy • Computer Vision

Tools & Cloud: Git/GitHub • Docker • Google Cloud • Azure • AWS • CI/CD • Expo • Vite

PROFESSIONAL EXPERIENCE

Undergraduate Teaching Assistant - Data Structures & Algorithms

VIT Chennai

Sep 2024 – Present

Chennai, India

- Mentored 50+ undergraduate students in DSA, conducting weekly tutorial sessions and code reviews
- Designed and evaluated assignments focusing on algorithmic problem-solving and optimization techniques
- Achieved 95% student satisfaction rate through personalized guidance and clear explanations

Google Student Ambassador

Google Student Ambassadors India

Oct 2025 – Present

Hybrid

- Selected from 10,000+ applicants to represent Google technologies across campus community
- Organized 5+ tech workshops and study jams reaching 200+ students on cloud computing and Android development
- Built strategic partnerships with student organizations to promote Google Developer Student Clubs

Founder & Full-Stack Developer

Sanchipt - AI-Powered News Summarizer

Jan 2025 – Present

Remote

- Architected and deployed ML-powered news aggregation platform serving 1,000+ daily active users
- Implemented NLP algorithms reducing reading time by 80% across 10 news categories
- Built scalable Next.js frontend with Python Flask backend, achieving 99.9% uptime and <2s page load time
- Integrated real-time news APIs and custom web scraping pipeline processing 500+ articles daily

Research Intern - Machine Learning

Centre for Cyber Physical Systems (CCPS), VIT Chennai

Jun 2024 – Jul 2024

Chennai, India

- Developed hybrid CNN-BiLSTM model with attention mechanisms achieving **92.73% accuracy** in skin lesion classification
- Published research in **Scientific Reports (Nature Portfolio, IF: 3.8)** outperforming InceptionV3 and VGG16
- Preprocessed and augmented HAM10000 dataset of 10,000+ dermatoscopic images using OpenCV and TensorFlow
- Collaborated with medical professionals to validate model predictions and ensure clinical applicability

Technical Lead - Web Development

Augmented & Virtual Reality Club VITC

Sep 2023 – Jul 2025

Chennai, India

- Led development of club website using Next.js and React.js, increasing engagement by 40% (500+ monthly visitors)
- Managed team of 15+ developers using Git/GitHub, conducting code reviews and implementing CI/CD pipelines
- Created 3D models and interactive AR experiences using Three.js, Unity, and WebXR technologies

KEY PROJECTS

NeuroDrive - Agentic Self-Healing EV Ecosystem

AWS • Agentic AI • Smart Mobility • IoT • Python • Cloud Architecture

2026

- **2nd Place - Tata Technologies InnoVent 2026** (2,822 teams, 10,247 innovators across India)
- Co-architected intelligent EV ecosystem with autonomous diagnostics and predictive maintenance capabilities
- Implemented agentic AI framework for self-healing vehicle systems reducing downtime by 65%
- Built scalable cloud infrastructure on AWS supporting real-time fleet monitoring and analytics
- Collaborated with team of 5 across November-January 2026, contributing to system design and AI integration

QueueHush - Smart Queue Management System [Play Store]

React Native • Firebase • QR Technology • TypeScript • Expo

2026

- Developed QR-based digital queue system eliminating physical waiting for 10+ partner businesses
- Implemented real-time position tracking with push notifications and Firebase Cloud Messaging

- Built privacy-first architecture with no user registration; currently in beta with 20+ test users
- Published on Google Play Store; received positive early user feedback on UX and usability

Instylo - All-in-One Social Platform

2025

[React](#) • [TypeScript](#) • [Appwrite](#) • [React Query](#) • [Jitsi Meet](#) • [Tailwind CSS](#)

- Architected full-stack social platform integrating 8+ features: networking, AI chatbot, video calls, marketplace
- Implemented real-time messaging with WebSocket, file sharing, and collaborative study rooms for 500+ users
- Built responsive UI with Tailwind CSS achieving 98% Lighthouse performance score and mobile-first design
- Integrated GPT-powered AI study assistant providing personalized learning recommendations

Nutrilytics - AI Nutrition Tracker [\[Play Store\]](#)

2025

[React Native](#) • [Firebase](#) • [ML Kit](#) • [TypeScript](#) • [Expo](#)

- Developed comprehensive nutrition tracking app with barcode scanning and offline-first architecture
- Integrated ML Kit for food recognition with 85% accuracy across 1,000+ common food items
- Published on Google Play Store

LiveScope - Real-Time Object Detection System

2024

[JavaScript](#) • [TensorFlow.js](#) • [COCO-SSD](#) • [WebRTC](#)

- **Winner - 1st Place, Scribble Pad Hackathon 2024** (100+ teams)
- Built real-time object detection system processing 30 FPS for surveillance and autonomous vehicle applications
- Implemented COCO-SSD model with custom training achieving 88% mAP on edge devices
- Optimized performance with Web Workers and WebGL acceleration reducing latency by 60%

ACHIEVEMENTS & PUBLICATIONS

Scientific Publication: "An attention based hybrid approach using CNN and BiLSTM for improved skin lesion classification" - *Scientific Reports (Nature Portfolio, IF: 3.8)* - May 2025

Patents Published: 3 patents in AI/ML and IoT - (1) Occupant Safety ML System (#202541116371), (2) Regenerative Power System (#20254110182), (3) Self-Sterilizing Article (#202541087249) - 2025

Competition Achievements:

- **2nd Place - Tata Technologies InnoVent 2026** (2,822 teams, 10,247 innovators nationally)
- 1st Place - Scribble Pad Hackathon 2024 | 3rd Place - Voyage Hackathon 2024 (1,000+ teams)
- 3rd Place - HalOthon Competition | Top 15 - HACKHAZARDS '25 (2,900+ teams, 5,700+ participants)

Professional Memberships: IEI Student Chapter Member - The Institution of Engineers (India) | Computer Science Division | VIT Chennai

Certifications: Google Cloud (TensorFlow, Dataplex) • Microsoft Azure (Static Web Apps) • HackerRank (SQL, Java) • NPTEL (Conservation Economics)